

4. Implement House price Prediction using appropriate machine learning algorithm.

CODE:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
# Sample dataset
data={
    'Year':[2015,2018,2012,2020,2017,2016,2019,2014],
    'Kilometers':[50000,30000,70000,15000,45000,60000,25000,80000],
    'Engine':[1200,1500,1000,1800,1400,1300,1600,1100],
    'Price':[500000,700000,350000,900000,600000,450000,800000,300000]
}
df=pd.DataFrame(data)
X=df[['Year','Kilometers','Engine']]
y=df['Price']
X_train,X_test,y_train,y_test=train_test_split(
    X,y,test_size=0.2,random_state=42)
model=LinearRegression()
model.fit(X_train,y_train)
pred=model.predict(X_test)
print("Predicted Prices:",pred)
print("Mean Squared Error:",mean_squared_error(y_test,pred))
# New Car Prediction
new_car=[[2022,10000,1500]]
price=model.predict(new_car)
print("\nPredicted Price:",price[0])
```

OUTPUT:

```
... Predicted Prices: [723071.60948653 481280.22104536]
Mean Squared Error: 755375696.472853

Predicted Price: 816555.3764678817
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning:
warnings.warn(
```